



The Dual Margins of Conservation: The political economy of rent creation and dissipation in fisheries management

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Outline

1. Problem statement



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2. Background Literature: Fisheries and Environmental Politics



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Problem statement

In fisheries economics we associate overfishing with rent dissipation. Efforts to restrict fishing effort and to achieve fisheries management target reference points often involve political lobbying of fisheries stakeholders including regional fisheries management organizations. From political economy and public choice theory, we know that political lobbying leads to rent dissipation. So, the question is whether the gain in rent from lobbying to achieve target reference points outweighs the loss in rent due to the costs of lobbying to achieve more sustainable fishing?



Background Literature: Fisheries and Environmental Politics

- ▶ Orach, K., Schluter, M. & Osterblom, H. (2017), 'Tracing a pathway to success: How competing interest groups influenced the 2013 EU common fisheries policy reform', *Environmental Science & Policy* 76, 90–102.
- ▶ Potter, B. (2002), 'Predatory politics: group interests and management of the commons', *Environmental Politics* 11(2), 73–94.
- ▶ Todd, E. & Ritchie, E. (2000), 'Environmental non-governmental organizations and the common fisheries policy', *Aquatic Conservation: Marine and Freshwater Ecosystems* 10(2), 141–149.
- ▶ Birchall, C. & Sanchirico, J. (2021), Lobbying over a dynamic resource: Evidence from a shared fishery, *Technical report*.
- ▶ Mills, E. N. (2023), 'The politics of transnational fishers' movements', *The Journal of Peasant Studies* 50(2), 665–690.



Background Literature: Public Choice

- ▶ Tullock, G. (1967), 'The welfare costs of tariffs, monopolies, and theft', *Economic inquiry* 5(3), 224–232.
- ▶ Tullock, G. et al. (1980), 'Efficient rent seeking', *Toward a theory of the rent-seeking society* 97, 112
- ▶ Skaperdas, S. (1996), 'Contest success functions', *Economic theory* 7(2), 283–290.
- ▶ Corchon, L. & Dahm, M. (2010), 'Foundations for contest success functions', *Economic Theory* 43(1), 81–98.
- ▶ Hillman, A. L. & Riley, J. G. (1989), 'Politically contestable rents and transfers', *Economics & Politics* 1(1), 17–39.



Basic Fisheries model

$$\frac{dX(t)}{dt} = rX(t) \left(1 - \frac{X(t)}{K} \right) - qX(t)E(t)$$

with no fishing $E(t) = 0$. So we have the initial value problem:

$$\frac{dX(t)}{dt} = rX(t) \left(1 - \frac{x(t)}{K} \right), x_0 = X(0)$$

The solution of which is well known: $X(t) = x(0) \frac{K}{x(0) + (K - x(0))e^{rt}}$

In steady state either $X(\infty) = 0$ or $X(\infty)$ approaches K in the limit.

So in the limit without fishing the unfished biomass will be K .



Rent

Rent is

$$\Pi = ph(E) - cE$$

Catch is

$$h(E) = qXE$$

In steady-state $X = K(1 - \frac{q}{r}E)$



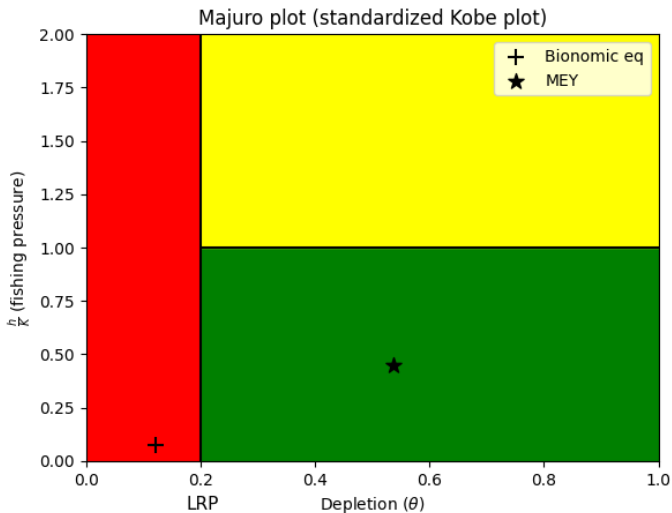
Reference points and bio-economic modelling

We can compare different reference points to assess whether the gains in rent exceed loss in rent from lobbying. Lobbyists such as NGOs and others should probably pick their battles.

- ▶ Compare Zero-rent to MEY (This is a net loss)
- ▶ Compare MEY and MSY
- ▶ Compare Pretty Good Yield and MEY



Honiara Plot: Comparing MEY and Zero-rent in a Majuro Plot



$$\Pi_B = 0$$

So

$$E\Pi_A = \frac{l_1}{l_1 + l_2} A$$

Substitute l_1^*, l_2^*



Does lobbying dissipate more rent than area based management with lobbying?

We consider two lobbyists competing for alternative prizes A and B

$$\max_{l_1} E\Pi_1 = \left(\frac{l_1}{l_1 + l_2}\right)A + \left(\frac{l_2}{l_1 + l_2}\right)B - l_1$$

$$\max_{l_2} E\Pi_2 = \left(\frac{l_1}{l_1 + l_2}\right)A + \left(\frac{l_2}{l_1 + l_2}\right)B - l_2$$

A and B are independent of the lobbying efforts l_1 and l_2 . Agent 1 might correspond to coastal state members of a regional fisheries management organization and Agent 2 might consist of either distant water fishing nations or alternatively international conservationist NGOs.



Solution

$$I^* = \frac{A - B}{4}$$

Assume $A = \Pi_A = pqKE_{MEY} - cE_{MEY}$ and $B = 0$ so that we are comparing maximum economic yield and the bionomic equilibrium.



Solution

Assuming A is the rent at maximum economic yield and B is the rent at the bionomic equilibrium (rent dissipation due to free entry)

$$I^* = \frac{r(Kpq - c)^2}{16Kpq^2}$$

This result will vary based on which reference points one wishes to compare.



Main result - Pessimistic

$$E\Pi_A(I^*, \Pi_{MEY}) - E\Pi_B(I_*, \Pi_\infty) = -\frac{3r(Kpq - c)^2}{16Kpq^2} < 0$$

Which demonstrates that lobbying to move from the free entry/open access solution to *MEY* dissipates more rent than is saved for a Tullock contest success function.



Multiple lobbyists and a single coastal state position

Conservation Lobbyists and coastal states lobby the RFMO

- ▶ Examples lobbying for harvest control rules vs no harvest control rule
- ▶ Labour standards
- ▶ limits on fishing effort
- ▶ BBNJ implementation



MEY vs Zero-rent

$$\max_{l_1} \Pi_3 = \frac{nl_1}{nl_1 + ml_2} A + \frac{ml_2}{nl_1 + ml_2} B - nl_1$$

$$\max_{l_2} \Pi_4 = \frac{nl_1}{nl_1 + ml_2} A + \frac{ml_2}{nl_1 + ml_2} B - ml_2$$

solving we get

$$l_1 = l_2 = l^* = \frac{m(A - B)}{(m + n)^2}$$

Assume as before is the rent at MEY and $B = 0$ is the bioeconomic equilibrium, then

$$l^* = \frac{mr(Kpq - c)^2}{4Kpq^2(m + n)^3}$$



Change in the probability of winning the contest

$$\frac{\partial I^*}{\partial n} = -\frac{mr(Kpq - c)^2}{2Kpq^2(m + n)^3}$$

or with $m = 1$ assuming an RFMO that has a consensus and n NGOs lobbying the RFMO and symmetric NGOs then

$$\frac{\partial I^*}{\partial n} = -\frac{r(Kpq - c)^2}{2Kpq^2(1 + n)^3}$$

So as NGO numbers increase the probability of achieving the new target falls. In the case examined here it wasn't really an RFMO but rather the great powers who had a consensus position lobbying for the freedom of the seas, and coastal states lobbying a law of the sea position.



Re-parameterized model

We reparameterize

$$\frac{dX(t)}{dt} = rX(t) \left(1 - \frac{X(t)}{K} \right), x_0 = X(0)$$

by setting $X(t) = \theta(t)K$.

$$\frac{d\theta(t)}{dt} = r\theta(t) (1 - \theta(t)), \theta_0 = \theta(0)$$

So the non-trivial steady-state is given by $\theta = 1$. Where $\theta(t)$ is the proportion of steady-state un-fished biomass. This facilitates comparison with fisheries management reference points.



Appendix: reparametrized model

$$\frac{d\theta}{dt} = r\theta(t)(1 - \theta(t)) - q\theta(t)E(t), \theta_0 = \theta(0)$$

In steady-state θ depends parametrically on fishing effort E :

$$\theta_{SS} = 1 - \frac{q}{r}E$$

or

$$E = \frac{r}{q}(1 - \theta)$$

and at MSY $\theta = \frac{1}{2}$.



Bio-economic models in a Majuro plot: The Honiara plot

The ordinate is $\frac{F}{F_{MSY}}$ or $h(X_\theta, E_\theta)/h(\frac{K}{2}, E(\frac{K}{2}))$

$$h(X_\theta, E_\theta)/h(\frac{K}{2}, E(\frac{K}{2}))$$

$$= \frac{2\theta E_\theta}{E_{\frac{K}{2}}}$$

$$= 4\theta(1 - \theta)$$

The abscissa in the Majuro plot is just θ in the Kobe plot it is $\frac{X}{X_{MSY}}$.



Conclusion

- ▶ In moving from one reference point to another actual change in rent depends not just on the benefits to the fishery but also on the cost to society of achieving that transition (political costs)
- ▶ The net change in rent is often negative
- ▶ We need to account for this when advocating for policy change.
- ▶ As fisheries managers tend to think in terms of Kobe and Majuro plots we should couch arguments in those terms and begin plotting fisheries economic models in their space
- ▶ For Majuro plots we should reparameterize the model in terms of the proportion of unfished biomass rather than the stock.



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Thanks for listening!

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